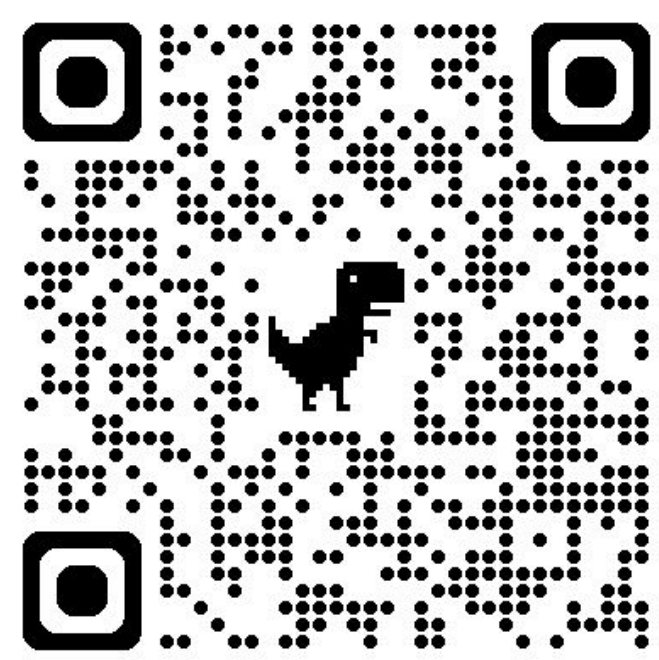


Scan the QR code for code, reference information and supplementary materials (to be updated).

Feel free to take a photo for later access.



Identifying Physical Activity Intensity Windows Associated with Joint Space Width Using Accelerometer Data

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Background & Introduction

Data Source:



Collected by the Osteoarthritis Initiative (OAI)

Large, longitudinal public health study

Data collected in the United States

Accelerometer Device:



Wearable accelerometer measurements

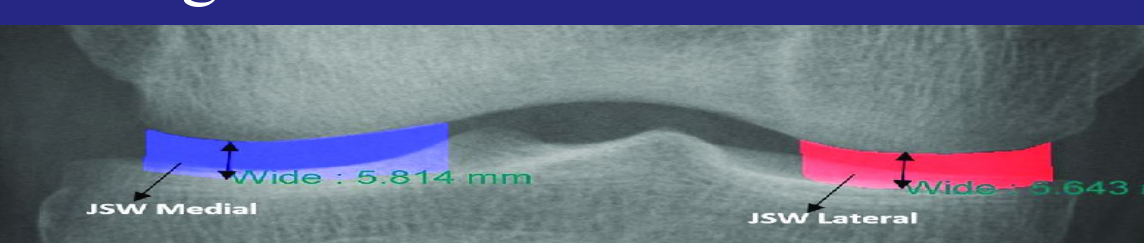
Minute-level physical activity counts

Multiple days per participant

Table 1: Variable Table of Raw Dataset		
Variable	Type	Description
V08PASStudyDay	Num	Study day (range 781–1427, with Day 1 = August 01, 2008)
V08MinSequence	Num	Time index (t): (range 1–1440, representing minutes from 00:00 to 23:59)
V08SuspectMinute	Num	Indicator of suspicious minute suggesting monitor wear during sleep
V08MINCnt	Num	The core variable, which is the activity count

Medial Minimum Joint Space Width (Outcome): measured in millimeters (mm), represents the minimum distance between the femur and tibia within the knee joint.

Figure 3: JSW Demonstration

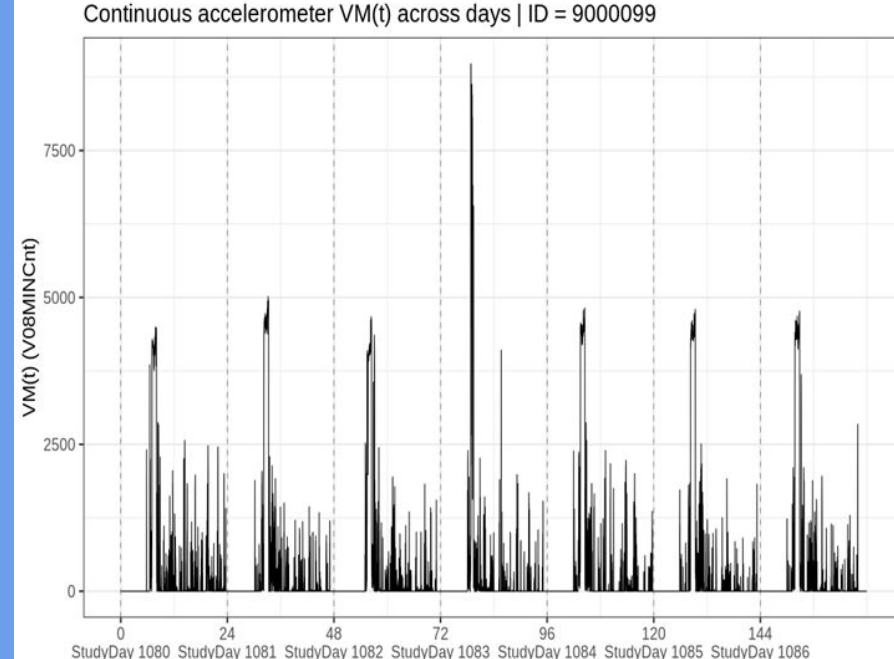


• A larger JSW: healthier cartilage and less structural degeneration.

• A smaller JSW: cartilage loss and narrowing of the joint space.

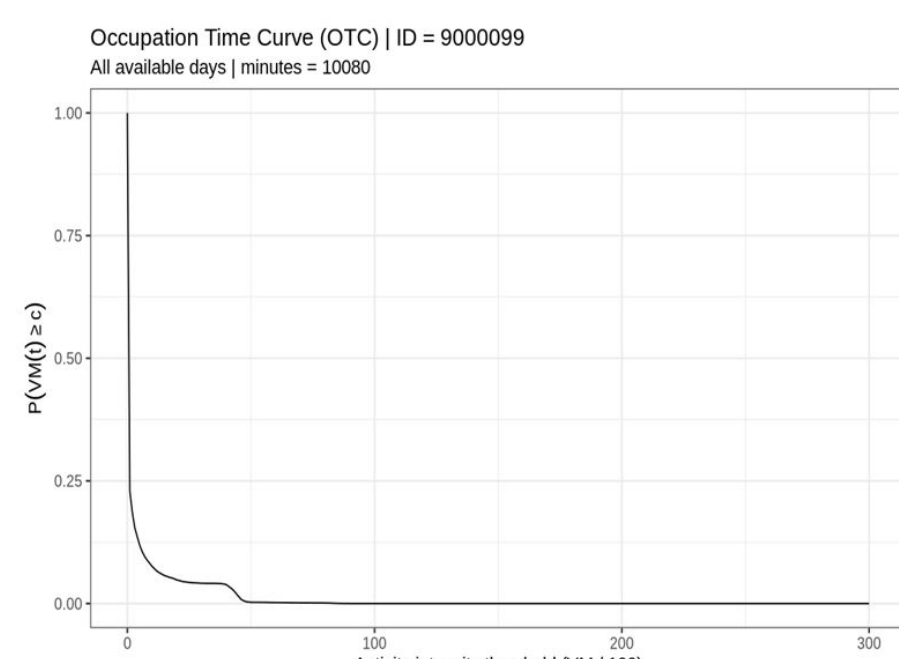
Data Preparation & Method Motivation

Figure 4: VM(t) Example



Continuous accelerometer VM(t) across days | ID = 9000099

Figure 5: OTC Example



Occupation Time Curve (OTC) | ID = 9000099

Figure 6: AUC Example

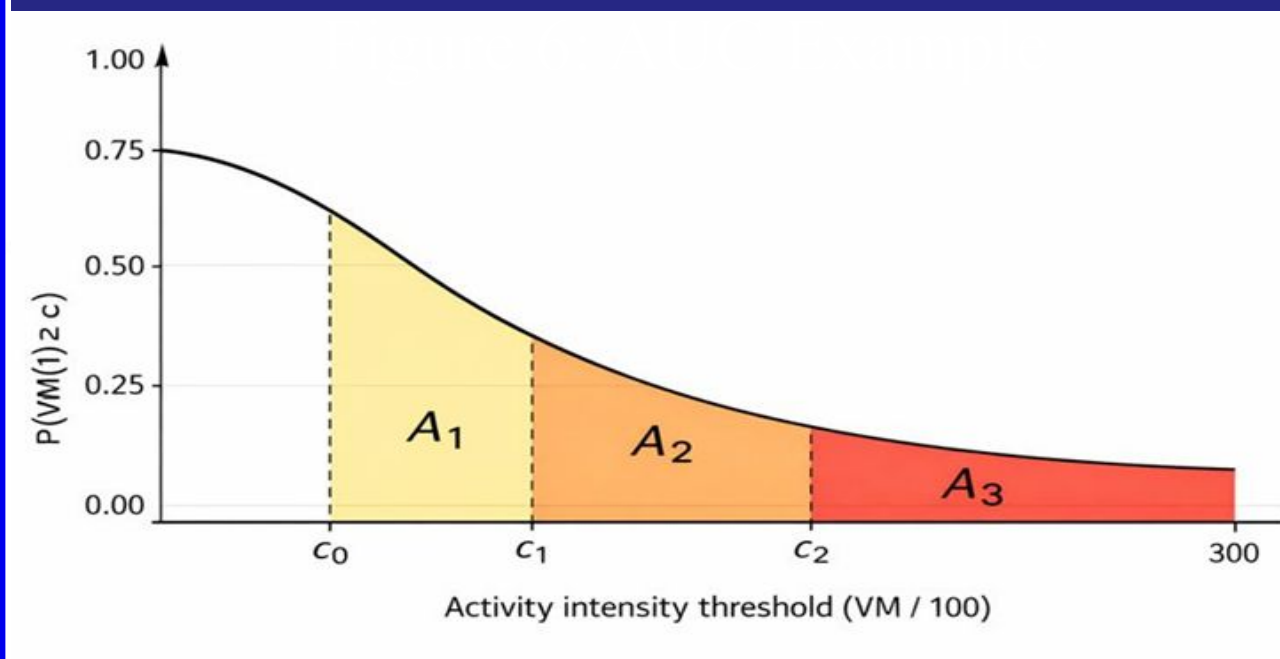


Figure 6: AUC Example

Figure 7: Sample Selection

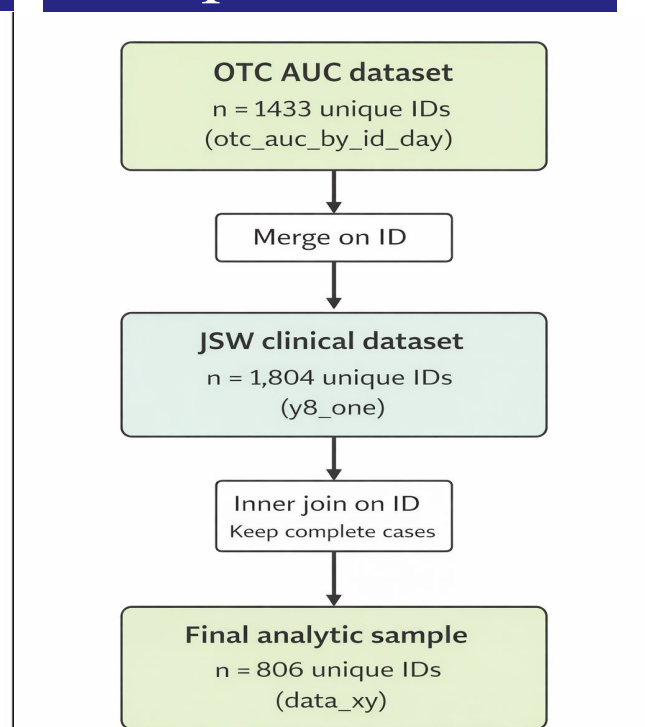


Figure 7: Sample Selection

We partition the activity intensity axis into intervals: $0 = c_0 < c_1 < \dots < c_J$, and define, for each interval, $A_j = \int_{c_{j-1}}^{c_j} OTC(c) dc$, $j = 1, \dots, J$.

By splitting the OTC into shaded areas, we convert a curve into a set of numerical features. OTC provides a stable functional summary. AUC converts it into a model-ready representation.

Vector Magnitude time series

VM(t): minute-level activity count at time t.

Figure 4 above shows activity intensity combining multiple days into a single continuous timeline.

Raw data: single time axis

This step: time re-indexing

From minute-level activity time series: $VM(t), t = 1, \dots, T$ to a distributional summary: Occupation Time Curve (OTC): $OTC(c) = P\{VM(t) \geq c\}$

The OTC intentionally ignores the temporal ordering of VM(t).

Table 2: Variable Table of Model-Ready Data

Variable	Description
Y_i	Outcome (Medial Minimum Joint Space Width)
Z_i	Additional covariate (currently not selected)
$X_i = (AUC_{i1}, AUC_{i2}, \dots, AUC_{iJ})$	Area under the activity curve across intensity intervals (Summary of time spent)

Methodology: Modeling Approach & Estimation Procedure

Functional Model Establishment

Ideal Model: $Y = \int \text{activity}(c) \times \text{effect}(c) dc$

This model directly captures how the effect varies with activity intensity.

Since the observed activity data are discrete rather than continuous,

Alternative Model: $Y_i = \sum_{j=1}^J X_{ij} \theta_j + \gamma Z_i + \epsilon_i$

Discrete approximation of a functional regression.

Due to high correlation across AUCs and high dimension,

Benefits of the Proposed Method

- Dimensionality reduction ($J \rightarrow K$)
- Improved stability by reducing multicollinearity
- Enhanced interpretability via segment-level effects

Optimization

Optimization formulation:

$$\min \sum_i \left(Y_i - \sum_{k=1}^K \beta_k \left(\sum_{j \in \text{segment } k} X_{ij} \right) - \gamma Z_i \right)^2$$

minimize the Residual Sum of Squares (RSS)

For each K, we estimate the corresponding segment specific coefficients under the segmentation constraints.

Model Selection

$BIC = n \log(RSS/n) + p \log(n)$

This balances model fit and model complexity:

A larger K allows more flexibility;

A smaller K leads to a simpler and more interpretable model.

Select the best segments K with smallest BIC.

Set: $\theta_j = \beta_k$ if $j \in \text{segment } k$

Convert pointwise effects into segment level effects.

Substitute the piecewise-constant assumption into the original model,

$$\sum_{j \in \text{segment } k} X_{ij} \theta_j = \beta_k \sum_{j \in \text{segment } k} X_{ij}$$

Final Model: $Y_i = \sum_{k=1}^K \beta_k \left(\sum_{j \in \text{segment } k} X_{ij} \right) + \gamma Z_i + \epsilon_i$

This stabilizes and enables interpretation of intensity ranges.

Why this model?

Figure 8: Correlation between AUCs

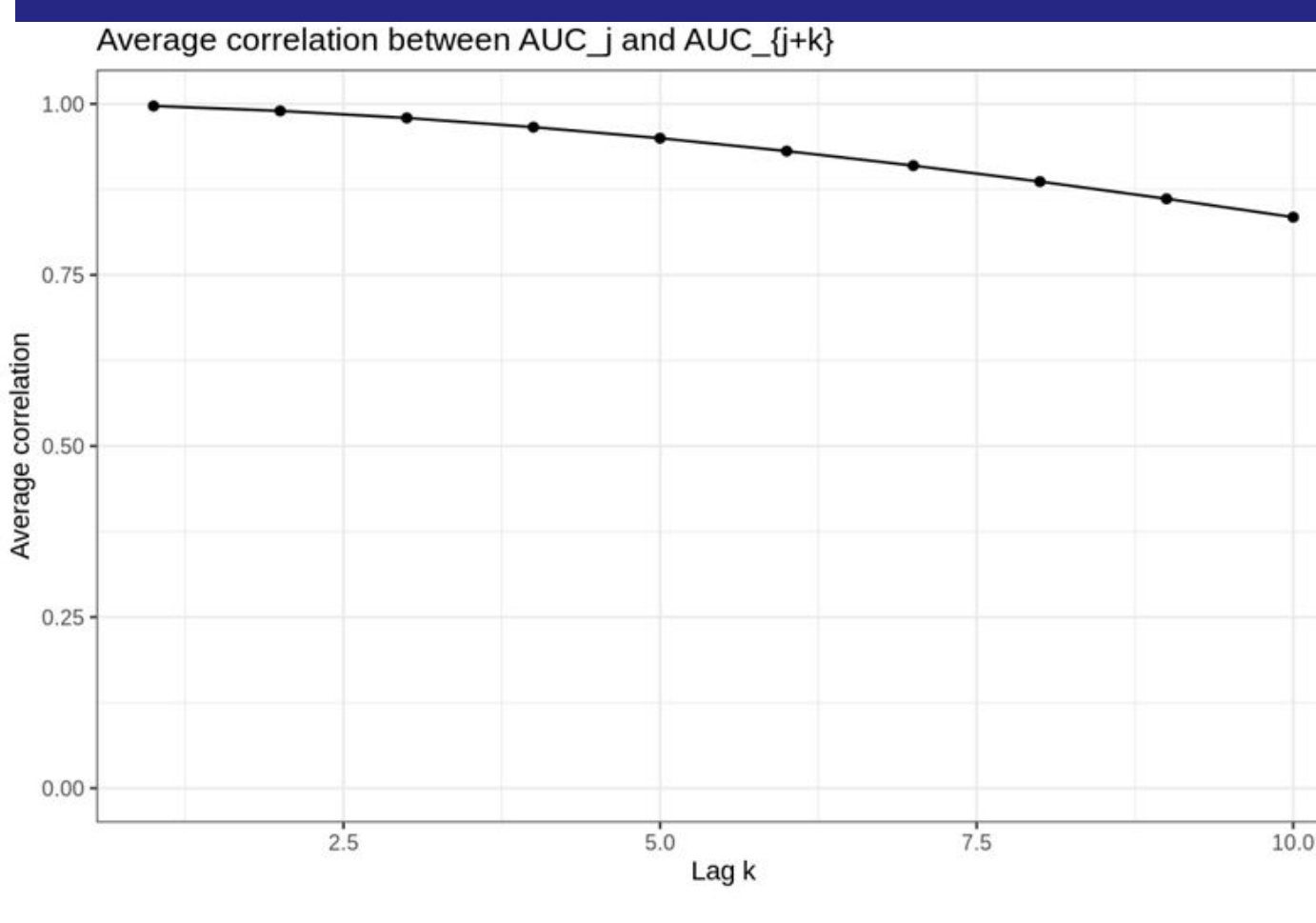
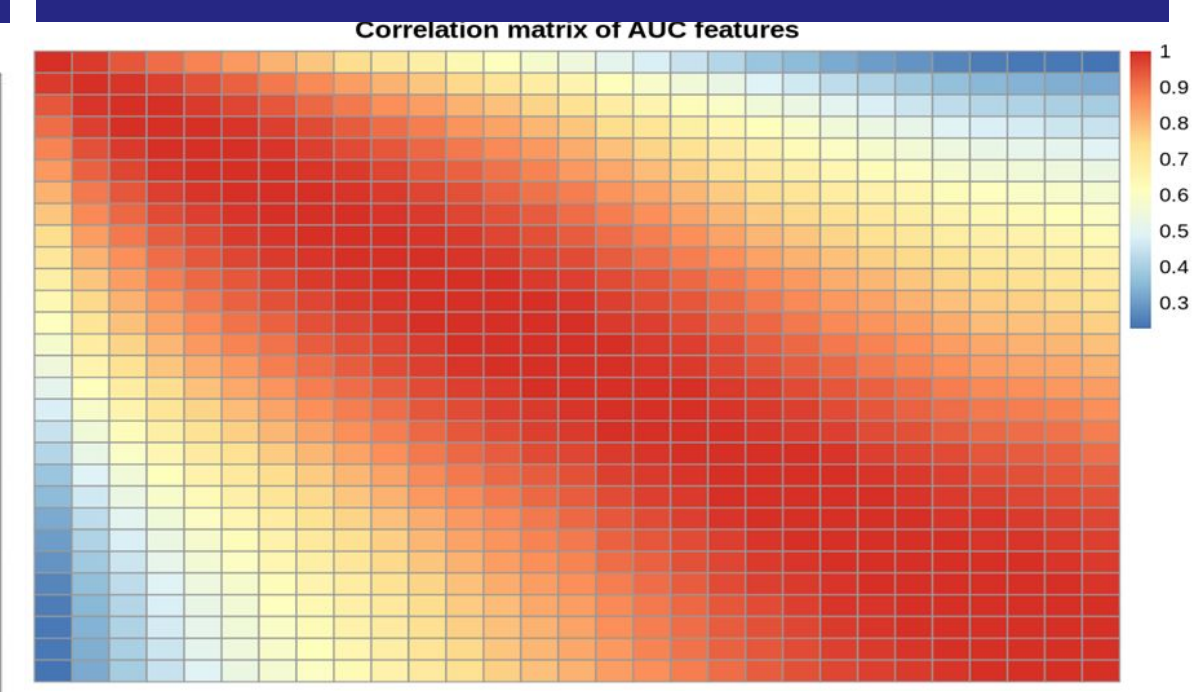


Figure 9: Correlation Heat Map



Strong correlation leads to applying fused lasso or standard linear regression unsuitable at this stage.

Result & Discussion

Figure 10: Model Result

Parameters	J = 60				J = 100				J = 200			
	K=1	K=2	K=3	K=4	K=1	K=2	K=3	K=4	K=1	K=2	K=3	K=4
β_1	0.0036	0.0171	0.0346	0.0344	0.0022	0.0105	0.0183	0.0090	0.0011	0.0052	0.0086	0.0044
β_2	—	-0.0002	0.0072	0.0077	—	-0.0001	0.0042	-0.2306	—	-0.0006	0.0021	-0.5324
β_3	—	—	-0.0023	-0.0366	—	—	-0.0013	0.2204	—	—	-0.0007	0.5254
β_4	—	—	—	0.0012	—	—	—	-0.0009	—	—	—	-0.0004
c_1 (jump)	—	11	3	3	—	18	6	28	—	37	13	58
c_2 (jump)	—	—	23	25	—	—	39	29	—	—	78	59
c_3 (jump)	—	—	—	27	—	—	—	30	—	—	—	60
γ	0	0	0	0	0	0	0	0	0	0	0	0
RSS	796.96	783.78	781.80	781.01	796.89	783.58	781.68	779.52	796.81	783.48	781.63	778.79
BIC	6.31	6.28	17.62	30.20	6.23	6.07	17.51	28.65	6.15	5.97	17.45	27.90

• The cutpoint: around 18% intensity range.

• Below this cutpoint: activity has a positively associated with the Joint Space Width.

• Above this cutpoint: association becomes slightly negative.

Result Validation: Permutation Test

Null hypothesis: The two-segment model is unnecessary: $\beta_2 = 0$.

$\Delta BIC_{obs} = BIC(K=1) - BIC(K=2)$ how K=2 improves the model over K=1 for observed data.

• shuffle JSW while keeping AUC fixed, creating a dataset with no association between activity and the outcome in each permutation.

$\Delta BIC_{perm}^{(b)} = BIC(K=1) - BIC(K=2)$ on the b-th permuted dataset.

Repeating this process B times produces a set of ΔBIC values that form the null distribution.

Indicator function: how often the permuted ΔBIC exceeds the observed ΔBIC .

The p-value is computed as the proportion of permuted datasets where the improvement in model fit (ΔBIC) is at least as large as that observed in the original data.

Table 3: Test Result

Permuted number	P-value
B = 50	0.0196
B = 80	0.0123
B = 100	0.0099

Improvement of K=2 over K=1 is statistically significant.

The null hypothesis is evidently rejected.

The permutation test supports the two-segment model structure.

Conclusion

Engaging in light to moderate activity is strongly associated with maintaining joint space, which reflects healthier cartilage and less structural degeneration.

Increasing activity intensity beyond the identified cutpoint does not appear to provide additional benefit for joint health.

Limitation

Observational data: no causal interpretation

Data-driven segmentation may vary across samples

The suspicious minutes are not considered at this stage

Need adjustment for confounders

Future Arrangement

Apply confounders (age, sex, BMI, baseline joint health)

Test cutpoint stability via bootstrap and subgroups

Handle suspicious activity patterns (non-wear, outliers)

Compare with alternative modeling approaches (e.g., models used by other group members)